

3 The global environmental crisis of transport

Patrick Moriarty and Damon Honnery

Introduction

Road vehicular travel is now in its second century. From almost zero vehicles on the world's roads in 1910, total vehicle numbers – those with four or more wheels – had grown to 1,003 million in 2007, of which 823 million – including sports utility vehicles in the US – were passenger vehicles (Altschuler et al., 1984: 2; Organization of the Petroleum Exporting Countries (OPEC), 2010: 81–90). By 2030, OPEC expects total vehicles to number over 1,700 million, including 1,370 million cars. US transport researchers Daniel Sperling and Deborah Gordon (2008: 4) also foresee similar vehicle numbers in 2030.

There is little doubt that such figures reflect the intentions of both the governments and citizens of the world, as discussed in Chapter 1. Populous countries like China and India, which presently have very low car ownership, wish to move to high-mobility societies, based on high ownership levels of locally manufactured vehicles. Given that most of the world's population still lives in countries with fewer than 25 cars per thousand population – compared with more than 400 per thousand population in OECD countries – there appears to be plenty of room for growth (Document 3.1 on this book's Companion Website expands on this theme).

At present, global vehicular travel is mainly constrained by incomes and, to a far lesser degree, by lack of road and parking space in some densely populated cities. In this chapter other possible constraints on future road travel are examined. Two are then selected – global oil depletion and global climate change from greenhouse emissions – as having crucial impacts on the prospects for future car travel. The various technical solutions that are proposed to overcome these constraints are then examined – namely improving the efficiency of transport vehicles and shifting to carbon neutral fuels – and we find that they will only be of marginal help. Given the limited value of such technological fixes, we then argue that a low-transport future is needed, and sketch out how this might occur. Finally this chapter discusses what urban transport might look like in the decades to come.

The global challenges facing road transport

Although this section will largely focus on the severe problems that oil depletion and global climate change present to transport worldwide, it will also examine three other challenges: road vehicle traffic casualties; air pollution from road vehicles; and the local environmental effects of oil extraction. Together, these are already widely acknowledged to be killing more than a million persons annually worldwide, with tens of millions more either seriously injured or made ill, and with disruption of societies and environments. Unlike the oil depletion and global warming issues, there is no disagreement about the basic facts. We have simply become used to this annual sacrifice for the doubtful benefits of high mobility. Even more importantly, only a small share of these high costs are being borne by the wealthy OECD countries.

Traffic casualties

Globally, road traffic fatalities in 2002 numbered over 1.2 million and a further estimated 50 million were injured (Ameratunga et al., 2006: 1533). The fatality rate per road vehicle varies inversely with vehicle ownership, so that, paradoxically, most fatalities are in countries with low car ownership, with Africa the continent with the highest death rate per vehicle. The explanation of this paradox is that low car-ownership countries tend to have very high numbers of unprotected road users – pedestrians and cyclists, and, especially in Asia, motorcyclists.

Consider a simple example in which all travel along roads is done either on foot or in cars. Assume that only 5 per cent of travel is presently by car, but that this doubles to 10 per cent. The number of pedestrians travelling will be reduced only marginally, but they will all now be exposed, *ceteris paribus*, to twice the traffic injury risk. Conversely, when most travel is by car or other four-wheeled vehicles, only a small percentage of travellers use the unprotected modes. The end result is that a high car-ownership country like the US, with around 25 per cent of the world's road vehicles, has under 4 per cent of global traffic fatalities (OPEC, 2010: 83; US Census Bureau, 2010: 844). Further, in nearly all OECD countries, the annual number of road fatalities has been falling in recent decades.

What will happen if global car ownership continues to rise, as is usually predicted? Given that vehicle ownership cannot be expected to rise much more in the high-mobility OECD countries – in fact, registered car numbers in the US fell by four million in 2009 over the previous year (Moriarty, 2010: 1) – the rise will come largely from higher ownership in presently low-mobility countries such as China and India. With the higher traffic fatality rate in low car ownership countries discussed above, global fatalities can be expected to rise disproportionately. Kopits and Cropper (2005: 176) studied the relationship between traffic fatalities and economic growth forecasts. They projected that global traffic fatalities would rise

about 66 per cent between 2000 and 2020, with a 30 per cent decrease in wealthy OECD countries being more than offset by an 83 per cent increase in poorer countries. Growth in poorer country fatalities was projected to continue until their per capita GDP reached US\$8,600 in 1985 values.

Air pollution

Air pollution is a major cause of mortality and sickness in cities. Traffic emissions are an important component of the urban air pollution problem, particularly in Asia, for two reasons. First, vehicle emission standards are often less stringent than they are in the industrialized countries of the OECD, and enforcement of existing standards may be less effective. Second, Asian cities have far greater population densities than Australian or North American cities (Newman and Kenworthy, 1999: 94), so many more people are affected by a given pollutant concentration.

This combination of a high density of both polluting vehicles and people in large cities can be a deadly one. In Asia, it was estimated in 2006 that outdoor air pollution was the cause of half a million deaths annually. Sixteen of the 20 most polluted cities in the world are in China, with vehicle emissions increasingly the main source of urban air pollution. Pollutants come not only from tailpipe nitrogen oxides, hydrocarbons and lead, but also from small particulate matter – suspended road dust, and particles from brake linings and tyres (Woodcock et al., 2007: 1081). Even in the UK, the urban residents at greatest risk from air pollution live in areas with the lowest car ownership. Although tailpipe emissions are anticipated to decline, increased hours per day in traffic, and the prevalence of cold start short trips – when emission control devices are least effective – will mean that adverse health effects will reduce much more slowly.

Environmental costs of oil extraction

As the Deep Horizon oil spill in the Gulf of Mexico very visibly demonstrated, there are heavy environmental costs in producing as well as combusting transport fuels. Because of progressive exhaustion of onshore oil reserves, attention is increasingly focused on the less accessible resources in deep offshore waters. In the US, as in other circum-polar nations, there is renewed interest in exploration of the Arctic Ocean. In the entire Arctic Ocean, reserves are estimated at 80 billion barrels of oil (Short and Murray, 2011: 162). For comparison, annual global production of oil in 2010 was 30 billion barrels of oil (BP, 2011: 8).

Short and Murray are blunt about the possible risks: ‘A large spill in the Arctic could not be contained or mitigated, and we should stop pretending otherwise.’ The nearest US Coastguard Station is 1500 kilometres away, and fog and snowstorms could hold up rescue work for weeks at a time. They further point out that:

each major marine oil production area in the United States has seen at least one catastrophic spill: the 1969 blowout of a drilling rig off the coast of Santa Barbara, California; the 1989 *Exxon Valdez* tanker spill in Alaska; and the 2010 Deepwater Horizon blowout in the Gulf of Mexico.

Ibid.

Exploration and production on land can also carry heavy ecological and social costs, particularly in countries with weak or poorly enforced environmental protection laws. In the head-waters of the Amazon ‘spills of oil, drilling mud and other toxic components associated with the exploration and exploitation of oil and gas have polluted large areas of rainforests, including streams and rivers’ (Jernelov, 2010: 360). As in the heavily polluted Niger delta in Nigeria, the local population receives few benefits from oil production, but must bear most of the costs of oil pollution (Moriarty and Honnery, 2008a: 869). Particularly in the Niger delta, the oil pipelines and other installations are often sabotaged, which along with illegal tapping of the pipes to siphon off products, further worsens pollution of the local creeks and land. The recently released report of the UN Environmental Programme reported, for example, that wells used by at least 10 Ogoni communities – the local people in the delta – had unsafe levels of hydrocarbons, and that the clean-up could take 30 years (Malakoff, 2011: 89).

Oil depletion

Despite much talk about biomass-based transport fuels and plug-in hybrids, road transport is still almost totally dependent on petroleum-based fuels. According to the International Energy Agency (IEA) more than 95 per cent of road transport energy is derived from petroleum, with nearly all the rest from natural gas and biofuels (IEA, 2011: 37).

The date at which ‘peak oil’ will occur is contentious, with all possible positions canvassed. Some argue that it has already occurred, others that it will occur in a few years, and still others that it is decades away, if it occurs at all. What is not disputed – although it is seldom discussed – is that on a global *per capita* basis, annual oil consumption peaked more than three decades ago in 1978 at about 0.7 tonnes, and from the early 1980s until the recent global financial crisis, was constant at close to 0.6 tonnes. Official bodies such as the US Energy Information Administration (EIA), the Intergovernmental Panel on Climate Change (IPCC), and OPEC are upbeat about oil availability, all assuming that the peak for oil production is decades off (for example, EIA, 2011: 231; OPEC, 2010: 63).

But when their forecasts for 2030 or later are expressed on a per capita basis, none of these organizations assumes that the per capita levels of the late 1970s will ever be reached again (Moriarty, 2010: 4). Nor do they expect any real rise over present levels, despite the low average level of oil consumption outside the OECD. It follows that if industrializing countries

like India and China wish to increase per capita use, other regions will have to reduce their consumption, given the expected growth in world population. Global oil consumption is rapidly becoming a 'zero sum' game. There are signs that at least some of the major consuming countries already realize this (Klare, 2008: 1). Further, it is too easy to forget that future generations could also require oil and other scarce resources for essential uses (Gardiner, 2010: 58).

Actual new discoveries of oil in recent years are also much more in line with the more pessimistic view – represented by the Association for the Study of Peak Oil and Gas (ASPO, 2009: 2) analysis – than the optimistic view – represented by the US Geological Survey (USGS). Given the unreliability of world reserve estimates discussed above, the more consistent US data will be used as an illustration. In a widely discussed 2000 report, the USGS predicted that US discoveries of oil, condensate and natural gas liquids over the 30 years from 1 January 1996 would have a mean value of 159 billion barrels (Petzet, 2000: 79). Yet over the first 15 years of this 30-year period, discoveries totalled only about 38 billion barrels (BP, 2011), rather than nearly 80 billion barrels expected if, conservatively, annual discoveries were spread evenly over the entire 30-year period. In a recent paper, Miller (2011: 1573) gives a good summary:

We agree that the world is not running out of oil, but the rate at which it can be extracted is falling. Oil will always be obtainable at a price, but that price may become unaffordable. Debates about whether the peak will be driven by inadequate supply, or by declining demand, are only different ways of stating the role of an affordable price.

Global climate change

Analysis of continuous cores painstakingly extracted from the Antarctic ice cap showed that the concentration of CO₂ in the Earth's atmosphere has varied from a minimum of 180 ppm (parts per million by volume) to around 300 ppm over the past 800,000 years. The lower values corresponded with glacial periods, and the high values with interglacials; calculated temperatures moved closely in step with CO₂ concentrations (Solomon et al., 2007: 24). By 2011, CO₂ levels were over 390 ppm, and are now rising at about 2 ppm annually.

As far as concentrations of CO₂ – and other greenhouse gases such as methane and nitrous oxide – are concerned, we are moving into uncharted territory. The concern is that Earth's climate will do likewise. The climate conference in Copenhagen in late 2009 (UN, 2009: 1) endorsed the European Union target of restricting the average global temperature rise since the start of the industrial era to 2 degrees Celsius (2 °C) in order to avoid 'dangerous anthropogenic climatic change'. The question of what is actually 'dangerous' and to whom or what is itself a controversial matter. A 2 °C rise might be survivable by the majority of humanity, though the

negative impacts will be unevenly distributed around the globe. But such a rise in temperature, if it occurs over the short time horizon of a hundred years or less, is likely to do great damage to the planet's biodiversity. Many species will not survive local habitat change and will become extinct. Whole ecosystems such as coral reefs and rainforests will be seriously threatened (Le Page, 2011a: 43; Warren et al., 2011: 150–54).

This cap on allowable temperature increase gives us very little leeway, since temperatures have already risen an estimated 0.76 °C, and we are committed to a further 0.6 °C rise because of the thermal inertia of the oceans. So, even if we ceased all emissions today, the Earth average temperature will eventually rise to 1.3–1.4 °C above the pre-industrial level. And given that more than 80 per cent of the world's energy needs are still met with fossil fuels, we also need to consider institutional inertia in emissions. We evidently have very little room to manoeuvre if we take the 2 °C limit seriously.

Of course, in a system as complex as the climate system, with its multiple feedbacks acting in different time frames, there are great uncertainties, both about the level of greenhouse gas concentrations that we can allow while keeping temperatures below the limit, and whether the limit itself is set too low. We can expect to see the limit steadily raised as we approach, and pass, the 2 °C limit. Nevertheless, if we assume this limit, then modelling shows that the world would have to cut CO₂ emissions in the year 2050 by as much as 85 per cent of their year 2000 value. This ambitious reduction target would mean that global annual fossil fuel use would have to be limited to about 50 EJ (EJ = exajoule = 10¹⁸ joules), rather than the 437 EJ we used in 2010 (BP, 2011: 41; Moriarty and Honnery, 2009: 34).

In 2009, Australian *per capita* emissions of CO₂ from fossil fuel combustion and cement manufacture were 17.9 tonnes, or 4.2 times the world average (IEA, 2011: 48). If emissions were to be reduced to 15 per cent of year 2000 levels, and equal for all the projected 9.15 billion humans in 2050, allowable emissions would only be 0.377 tonnes per capita (BP, 2011: workbook). Australian 2009 CO₂ emissions would need to be cut by a factor of 47. Cuts in US emissions would be similar, but less for other OECD countries, typically 25–30 fold. Put simply, our emissions would need to be cut to near zero. At 0.377 tonnes per capita, even industrializing countries such as China, India, and Indonesia would see large reductions in CO₂ emissions (IEA, 2011: 48).

One prominent climate researcher, James Hansen, argues strongly that even a 2 °C limit, though very difficult to achieve, is itself too high. In a recent paper (Hansen et al., 2011: 5), he claimed that the paleoclimatic data point to a limit of only 1 °C above pre-industrial if irreversible destabilization of the polar ice-caps is to be avoided. Or as Michael Le Page (2011b: 41) put it: 'In comparable interglacials in the past half million years, when temperatures were less than 1 °C warmer than they are now, sea level was around 5 metres higher.' As we are already committed to exceeding this temperature rise, it follows that even the existing CO₂ level

of over 390 ppm is too high, and that it should be *reduced* to 350 ppm. Not only would annual emissions have to drop from their 2010 level of over 33 billion tonnes from fossil fuels alone (BP, 2011: workbook), but some past emissions would also have to be removed from the atmosphere. Of course, in our present political climate, such cuts are wildly infeasible. It is all too easy implicitly to think that because an action is politically infeasible, the problem will disappear. Unfortunately, the natural world doesn't work that way.

Regardless of the CO₂ levels that are agreed, transport will not be exempt from these emission cuts. On the contrary, it is likely that urban passenger transport will have to make larger reductions than the economy overall. In the transition to far lower oil use and emissions, freight transport can make claims for priority in delivering food to cities and raw materials and intermediate products to industry. Without freight, industrial economies would soon collapse. Cities in the West have only three days' food supply, and so must be continuously resupplied. Even non-urban travel could claim priority over urban travel, given that alternative modes are more difficult to organize, especially for remote settlements. Accordingly, the remainder of this chapter will examine the implications for urban passenger transport, mainly in an Australian context.

Proposed solutions: technological fixes

Possible technological fixes include improving the fuel efficiency of conventional road vehicles, both passenger and freight, greatly expanding the use of alternatives to petroleum fuels, and the development of new propulsion systems. Although none of these helps reduce traffic casualties, they do promise simultaneously to tackle oil depletion and air pollution, both local and global – from CO₂ and other greenhouse gases.

Improving vehicular energy efficiency

The energy efficiency of a vehicle is usually measured by comparing the vehicle-kilometres of travel for a given fuel input, as in the familiar 'miles per gallon', or in today's units, kilometres per litre of fuel. The best way to think about energy efficiency is to look at what the energy in the vehicle's fuel tank is actually used for. Some is used for lighting, air-conditioning, powering devices such as windscreen wipers, vehicular entertainment and information systems, and for power steering, braking, etc. But most is used to overcome a vehicle's resistance to forward motion. This resistance, or *road load*, has three components. The first, rolling resistance, is the frictional resistance between tyres and the road. This resistance can be reduced by better tyre design and higher tyre pressures – but higher pressures give a less comfortable ride.

The second component is air resistance, which is important at the higher speeds attained in highway and freeway driving. It can be reduced by better

attention to aerodynamic design – and, of course, by reducing driving speeds. The third component is the most important for inner city driving: overcoming vehicle inertia, that is, getting the vehicle to a higher speed from either a stationary or a lower speed state. Like rolling resistance, inertial resistance varies directly with vehicle mass, so reducing the loaded mass of the vehicle can reduce inertial resistance. Smaller vehicles are thus more fuel efficient than large ones. Cutting down the amount of stop-start driving will also reduce the road load.

All the energy supplied to the wheels of a road vehicle – whether car, truck, or bus – is used to overcome this road load, so reducing it will increase vehicle fuel efficiency. But another approach is to improve the engine and drive-train efficiency, that is, increasing the share of the input fuel energy in the tank that results in propulsive energy at the wheels. Over the past century, engine efficiency has been continuously improved, but has resulted in little overall improvement in kilometres per litre. The reason is that the energy demands on the vehicle have also risen, negating the engine efficiency progress. The road load has risen, mainly because of the shift to larger, heavier vehicles. So has the energy needed to run the auxiliary equipment: air conditioning, power steering and the like.

The overall result is that there has been little improvement in fuel efficiency worldwide. Woodcock et al. (2007: 1085) reported that present-day Ford cars in the US were no more efficient than the model-T Ford. For the Australian car fleet, our own analysis showed that over the two decades 1985 to 2005 there was no gain in fuel efficiency (Moriarty and Honnery, 2007: 5).

Can we do better? As could be inferred from our discussion, maximum efficiency gains will come about from combining both drive-train efficiency gains with reductions in road load. Inertial resistance dominates road load under stop-start urban driving conditions. A shift to electric drive vehicles, as in the hybrid vehicles presently being sold – such as the Toyota Prius – promises to reduce road load by regenerative braking. Here, instead of the kinetic energy of the moving vehicle being converted to heat and dissipated by the brakes, it is converted into electrical energy and stored in batteries for re-use later (Moriarty and Honnery, 2008b: 1720). Also, because hybrids have both an electric drive motor and an internal combustion (IC) engine, it is possible to use a smaller-sized – and so lighter – IC engine, and run it under more constant load conditions. The result is that hybrids can be much more efficient than conventional IC vehicles under urban driving conditions.

If both all feasible road load reductions and hybrid drive-trains were implemented on future vehicles, a Swedish review (Akerman and Hojer, 2006: 1946) estimated that a four-fold rise in fleet fuel efficiency for urban driving might be possible by 2050, and a threefold rise for highway driving. By 2030, it might, at best, be possible to double car fleet fuel efficiency (Moriarty and Honnery, 2008b: 1722). Compared with the dismal history of fuel efficiency gains, a three- or four-fold rise seems

very optimistic. But when compared with what is needed for climate change mitigation by 2050, even improving efficiency by a factor of four is nowhere near enough.

Two additional factors must be considered. Since presumably the aim is to move people, not vehicles, we should really consider passenger-kilometres per litre of fuel, not vehicle-kilometres per litre. Urban car occupancy rates are typically about 1.5 or less today, so there is much scope for improvement in typical five-seater cars. But in Australia and other high-mobility countries, occupancy rates are falling, and we will be doing well if we can prevent further falls in the face of declining household sizes and rising car ownership (Moriarty and Honnery, 2012: 1963). The other factor concerns the energy required to get a litre of fuel to the fuel tank. As conventional oil is depleted, the world will increasingly have to rely on unconventional oil in future. But deepwater oil, and oil from tar sands or shale oil deposits have far higher energy needs and greenhouse gas (GHG) emissions per unit of fuel delivered to the fuel tank. Like falling occupancy rates, these higher energy needs for fuel production will counteract any hard-won improvements in vehicle fuel efficiency.

Shifting to alternative fuels/power systems

If we cannot expect much in the way of reductions in either oil use or GHG emissions from improving vehicular fuel efficiency, what about changing the fuel? Not only will oil be expensive and in more restricted supply in future, but, as we have just seen, its GHG emissions per litre of delivered fuel, already a problem, will get worse. Moving to clean – or ‘green’ – fuels is an obvious answer. Such an approach promises to solve the oil depletion problem, and both local and global pollution.

The use of non-petroleum fuel for transport is growing, with the leading countries Brazil and the US. In Brazil ethanol is made from sugar cane, and in the US from corn – and sorghum. In the EU, although some ethanol is made from grain, the preferred alternative fuel is biodiesel, made from edible oil seeds such as rapeseed. Global production is minor compared with ethanol. Worldwide, liquid biofuels production in 2010 was the equivalent of 69.1 GL (gigalitres) of oil, or about 2.8 per cent of all refined transport fuel including non-road transport (BP, 2011: 39; IEA, 2011: 39). Since all transport biofuels are presently based on food crops, it is recognized by all that further expansion of biofuels will soon need a move to cellulosic ethanol.

Liquid transport fuels based on edible crops not only raise ethical questions, but are only of limited help in reducing transport GHG emissions. Although US corn ethanol is often reported as giving roughly a 20 per cent – not 100 per cent – saving in carbon emissions, two factors not usually considered actually negate any climate change advantage, even if it does replace imported oil. First, corn production in the US requires nitrogenous fertilizer, and emissions of nitrous oxide – an effective and long-lived

greenhouse gas – from the soil alone negate any carbon emission savings (Crutzen et al., 2008: 389).

Second, as Searchinger et al. (2008: 1238) have pointed out, there are flow-on effects in food-importing countries from using US corn for fuel rather than food. When the global consequences are considered, the authors again found that corn ethanol produced higher GHG emissions than petrol. Using cellulosic ethanol from switchgrass grown on corn land was also argued to produce higher emissions. Because of the system considerations discussed here, it is likely that cellulosic ethanol – which is not yet in commercial production anywhere – from biomass plantations, or even from agricultural and forestry wastes, will produce little or no net GHG emission savings, as Felix and Tilley (2009: 410) also argue.

As discussed above, electric drive is an efficient means of propulsion because it allows for regenerative braking. A further advantage from a climate change mitigation viewpoint is that the propulsive electrical energy needed can be derived from a variety of sources, including renewable energy (RE). In fact the primary output for many RE sources – wind, solar photovoltaic, hydro – is electricity. At present, the only road vehicles with electric drive are hybrid electric vehicles, but the electric power produced is ultimately derived from the petrol tank. Future vehicles, however, could be powered by mains electricity.

Several options are possible. The most popular one at present is plug-in hybrids, which would have two energy sources: a conventional petrol tank, and a battery pack charged overnight from mains electricity. It is also possible to have a fully electric vehicle, without a petrol IC engine, but the battery requirements would then be excessive – and very expensive. For plug-in hybrids, the idea is to size the battery so that the vehicle can go, say, 40 kilometres without recharging. For days with greater travel needs, the motorist could then use the petrol engine.

Developing a battery with sufficient energy density – energy stored per unit mass of battery – has been the main difficulty for both fully electric and even plug-in electric vehicles. The batteries must not only have high energy and power density, but must be made from non-toxic materials that are available in adequate supply, be safe to use, be able to withstand thousands of charge/discharge cycles, have low energy conversion losses, and be not too expensive. No battery will ever come near the energy density of petroleum fuels, but batteries based on the lightest metal, lithium, are being promoted as the answer for plug-in hybrids. As discussed in Moriarty (2010: 9), lithium-based batteries are not only very expensive, adding as much as US\$18,000 to the cost of a vehicle, but, at least according to OPEC, also could not support more than a small fraction of the large projected vehicle fleet discussed above in the first section because of the limited lithium resource base. Others dispute this lack of lithium reserves. The question may not even matter, since the high cost and safety problems of lithium-based batteries means that research on other battery types continues.

A more radical approach would use hydrogen as a fuel in hydrogen fuel cell vehicles (HFCVs). The hydrogen, if derived from electrolysis of water, could eventually be based on electricity from renewable energy (RE), but at present hydrogen is produced from natural gas. A large number of difficulties are yet to be overcome for HFCVs to be a viable option, including the high cost but limited life of the fuel cell pack, and limited availability of the platinum used for catalysing the reactions at room temperature. But just as serious is the difficulty of storing hydrogen onboard in cars, although this is far less of a problem for large vehicles, including trucks, buses and trains. Not only are there serious safety and other obstacles with storing hydrogen as either a highly compressed gas or a very low temperature liquid, but the energy costs of compression/liquefaction and transport to refuelling stations will be high (Bossel, 2006: 1829). Using hydrogen directly as a fuel in IC engine vehicles would face the same problems of hydrogen onboard storage and transport.

Although, like any alternative fuel, hydrogen would help overcome the oil supply problem, it would only be advantageous from a GHG viewpoint if made from RE electricity sources. Yet a marked shift to RE for electricity will not happen anytime soon. At present, the share of fossil fuel in global – and Australian – electricity generation is still rising slowly, and official forecasts are that this situation will change little over the next two to three decades (EIA, 2011: 254). And until such time as carbon-neutral energy accounts for almost 100 per cent of electricity generated in a grid – as is true only for some small countries like Norway and Iceland – GHG production in the economy overall will always be less if such electricity is used *directly* in end-use applications. So applications such as for traction power for electric public transport, or for electronic equipment should have priority over battery or hydrogen-powered vehicles, where a large fraction of the generated carbon-neutral electricity will be lost in conversion, for example to hydrogen.

To summarize, neither vehicle energy efficiency nor alternative fuels looks likely to offer dramatic reductions in either oil use or GHG emissions in particular. Even on a passenger-kilometre basis, it will be difficult to achieve much in the way of reductions because of declining vehicle occupancy rates, declining shares of carbon-neutral energy for electricity generation, and increased reliance on non-conventional oil. Population increase will further offset reductions on an absolute basis. For Australia, at best a threefold reduction in passenger car GHG emissions per kilometre might be possible by 2050, but, as we have seen, a 50-fold reduction might be needed. As Rees (2009: 300) has argued, putting our faith in technological fixes for transport problems promises to make matters worse by deflecting attention from the changes really needed.

Proposed solutions: social changes

Most discussions of the need to reduce either transport oil use or GHG emissions assume that only technology-based solutions are possible. On the

other hand, reductions based on social changes are not rated highly (see for example, the IPCC in Kahn Ribeiro et al., 2007: 360). Infinite possibilities for technological change are assumed, but very limited possibilities for social change – except for those needed to adjust to the new technology. Although such a position closely matches corporate interests, the preceding section showed that technological fixes will not give us anywhere near the needed cuts in either oil or GHG emissions.

The changes needed are of two kinds. First, we will need to reduce our levels of vehicular travel, as measured by passenger-kilometres per capita. Second, we will need to move nearly all the remaining vehicular travel to public transport. Many people seem to believe that shifting to public transport will be enough, because it is believed to be vastly more efficient than car travel. But on a primary energy per seat-kilometre basis, the public transport modes are typically only about two to three times as efficient as car travel (Moriarty and Honnery, 2011: 192).

However, several other factors need to be considered, and they all act to shift the balance towards public transport. The occupancy rate – the proportion of seats filled – for cars in OECD countries is today typically about 30 per cent, or 1.5 persons per 5-seater car. Public transport occupancy rates in Australia, though rising, may not be much higher than this at present, but they can potentially be much higher – as in Tokyo. There is, of course, a downside to all this: the transport planner's high occupancy rate is the commuter's grossly overcrowded peak hour train.

Another undoubted benefit of electric public transport is its independence from oil. It is likely that urgent action will be needed to curb oil imports before serious emission reductions are imposed to mitigate global warming.

The most important – and least appreciated – factor is that total vehicular passenger travel levels are not independent of the dominant travel mode in a city. Several lines of evidence support this conclusion. First, no city in the world with public transport as the dominant mode has ever approached 5000 passenger-kilometres per capita of public transport, except Tokyo, with 5500 in 1990 (Newman and Kenworthy, 1989: 342, 1999: 84). One obvious objection is that public transport-dominated cities have either been low-income cities – for example, Australian cities in the 1940s and 1950s – or very densely populated, or both, as was the case for Japanese and other Asian cities up to the 1960s. To answer this objection, we first need to consider travel times for different modes.

Public transport modes are inevitably slower on a door-to-door basis than car travel, except, perhaps, for trips to or near the city centre. Reasons for this include the need for public transport to stop frequently to drop off and pick up passengers, the time delay for any trip interconnections, and the circuitry of many non-central public transport trips.

Passengers are conventionally regarded as considering two costs of travel in their decision making: the monetary costs of travel, and the time costs of travel (Moriarty, 2002a: 3). Car travel may be more expensive on a fully monetary costed basis than public transport – although the *perceived* costs

for any given trip may be less – but the time costs will be far higher for most trips if undertaken by public transport, at least as it presently exists in most cities.

As urban populations become wealthier, more and more of them will become time-constrained rather than money-constrained, and will prefer the faster mode. So even given relatively cheap travel, if only slow travel modes are available, they will travel much less. This time factor, rather than income, largely explains why travel levels in cities with high public transport use will often be lower than in car-oriented cities. Paradoxically, a further consequence may be that *overall* travel times are less in public transport-oriented cities. Such appears to have been the case for Melbourne as it shifted from public to private travel (Moriarty, 2002a: 5). To summarize, travel costs have both a money and time component; if either is reduced, people will travel more.

Third, although travel is usually seen as a derived demand, not all travel is undertaken solely for utilitarian purposes. For many trips, it is truly ‘better to travel than to arrive’. Particularly for young males, and even for others, mastery of a car provides psychic – i.e. non-instrumental – benefits. Even for the commuting trip, many drivers see it as a time when they can be alone – ‘quality time’ in today’s jargon. Kaufmann (2005: 50–51) showed from his European data that the way people value travel – by different modes – varies considerably. He identified four types of travel valuation behaviour, which he defined as follows:

- *Efficient*: people who want to get from trip origin to destination in the shortest possible time.
- *Sensorial*: people who value the experience of the travel for its own sake – for example, walking or cycling – related to experiencing the environment travelled through.
- *Pre-planning*: people who minutely determine travel in advance, in which travel is integrated with valued activities – working or relaxing on the train, music in the car.
- *Flexible*: people who react to opportunities that present themselves in the course of travel.

Fourth, not only trip times for any given trip, but also tripmaking patterns will vary with the dominant mode. At present in the car-oriented Australian cities, car travel calls all the shots. Since around 90 per cent of vehicular travel is by car, most destinations are now those than can be easily accessed by car, not alternative modes. Travellers using other modes have to fit in as best they can. What this means is that the inevitable difference in door-to-door trip times is exaggerated. If the slower public transport modes were dominant, there would be far more incentive to combine trips for different purposes. At present the speed of car travel allows activities to be *interwoven*; parents may chauffeur their children to Saturday morning sport (as discussed in Chapter 10), then go and do some shopping before returning

to pick them up. Because of time constraints, such interweaving may simply not be possible for public transport – and for older children, chauffeuring would simply not be done.

An important question arises: Can the existing car-oriented cities of Australia and elsewhere be adapted for travel by alternative modes? Will this prove impossible without massive restructuring of the urban form? We do not think so. In Chapter 9 the means of such adaptation are proposed. A look at the changing urban structure of Australian cities supports our contention that adaptation is possible. At the time of Australia's first post-war census in 1947, the capital cities were highly centralized. Most jobs and retail sales were usually in the municipality containing the city's CBD, and the electric tram and rail systems were oriented towards the city centre. Up to 80 per cent of vehicular travel in capital cities was by public transport, mainly trains and trams (Moriarty and Honnery, 2005: 2). With both destinations and dominant travel modes oriented towards the city centre (as also noted in Chapter 2), the result was that most vehicular trips were similarly oriented. Average vehicular trip lengths were thus strongly influenced by the average resident distance from the CBD, which was naturally higher in bigger cities.

In contrast, today's cities have a much more even spatial balance of residences, workplaces, shopping and leisure centres, and other services. So although vehicular travel per capita in Melbourne, for example, is today about four times its year 1947 value, travel *needs*, as measured by the separation of residences from jobs, shops and other destinations, have actually reduced (Moriarty and Honnery, 2008a: 870). What this suggests is that although car-oriented cities have far greater travel levels, this is a result of the higher travel *convenience* and faster door-to-door speeds of car travel. But the type of urban form tomorrow's public transport systems will inherit are *potentially* low-travel cities. Further, the intermixing of land use functions makes walking and cycling possible for many trips.

A possible future for urban transport

We will now try to give some idea of what shape passenger travel in Australian cities will likely take in the coming decades. First, it must be recognized that any serious attempt to tackle global warming will require reductions in *all* sectors of the economy, not just transport. At first glance, oil depletion seems mainly a problem for the transport sector, given its heavy reliance on petroleum products. However, global oil depletion will have significant effects on the global economy more broadly (ASPO, 2009: 3). Change in all economic sectors means that the nature of work, the type of jobs people do, will also change.

Particularly if economies worldwide become more oriented towards satisfying the human needs of all the world's people (Moriarty and Honnery, 2011: 187), it is difficult to see much future in Australian cities for today's heavy emphasis on banking, finance, real estate, marketing, and the

obsession with retail sales. This type of employment is concentrated in the centres of our large cities, and the loss of many of these jobs will mean a loss of public transport patronage from these job-holders. Other functions of the city centre will survive, and new types of employment can be expected to develop. The centre will remain the most accessible region of the city, and is the logical place for the location of major sporting, entertainment and education centres, city – or even State-wide – administration offices, specialist shops and services. Increased public transport travel to these destinations will offset much of the patronage loss just discussed.

It is also important to recognize that future urban transport systems will have to take the urban form largely as given; because buildings and urban infrastructure have a lifetime of many decades, there is not the time for extensive modifications. Many writers have seen urban densification as a key factor in making urban transport more ecologically sustainable. Even if such densification helped urban transport, and there are doubts about whether such changes are necessary, it might not help – and could even hinder – other components of urban sustainability. Future cities, for example, may need to be more self-sufficient in food – mainly vegetables and fruits, not cereals – water and even energy provision. The existing lower density settlement patterns of large Australian cities could even help here. What is optimal from a sustainable transport perspective may conflict with sustainability in other sectors; compromises may be necessary. Even so, it makes sense to have higher residential and workplace densities in the central area and around public transport nodes such as railway stations. It is even possible that the large metropolitan centres could lose some population to rural areas, given the growing demand for food globally. If this happens, it would further slow the pace of urban change.

The changes to employment, or, more generally, to a ‘human needs economy’ will not happen rapidly. Since we do not really know what the urban future will be, we will have to feel our way. Although large-scale change to the existing urban fabric is unlikely for the reasons just given, there should be enough time to implement many ‘micro-changes’ in the way it is used. For example, local shopping centres could be used much more intensively than at present, with accordingly less use of drive-in shopping centres. Many existing workplace buildings could be used for the new job types that will inevitably arise in ecologically sustainable cities.

Australia’s cities of the future will likely be low-mobility cities, a requirement that follows from energy and environment constraints. Given these constraints, much more *localization* of activities is probable – working, socializing and entertainment, shopping and so on (Moriarty and Honnery, 2008a: 871). Non-motorized transport is ideally suited to these local trips. For longer trips, bus travel could be more important than today because, in the transition, rapid changes and new combinations of bus routes are possible because of the vast networks of urban roads – our inheritance from the automobile age.

